Riko Al Tayer

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EDUCATION

University of Cambridge | *M.S. Computer Science* 2015 - 2017

- Specialized in distributed systems and concurrent programming, thesis on lock-free data structures improving throughput by 34%
- Led computer science society with 200+ members and organized weekly technical seminars on microservices architecture

Imperial College London | *B.S. Mathematics with Computer Science* 2011 - 2015

- Graduated with First Class Honours; awarded Best Algorithm Design Project for optimizing search queries by 67%
- President of Coding Society, organized annual programming competition attracting 300+ participants

EXPERIENCE

Stripe | *Senior Backend Engineer, Payments Infrastructure* 2021 - Present

- Architected distributed payment processing pipeline using Kafka and gRPC, reducing latency by 42% and enabling 500K+ transactions per second
- Led migration of legacy monolith to microservices for fraud detection system, decreasing detection time from 8s to 280ms and improving accuracy by 28%
- Mentored 4 junior engineers on system design and Go best practices, resulting in 3 internal promotions

Cloudflare | *Senior Software Engineer, Edge Computing* 2019 - 2021

- Designed and implemented edge caching layer handling 12 billion daily requests, reducing origin load by 78% and saving $2.1M annually
- Optimized PostgreSQL queries and implemented Redis caching strategy, improving API response times from 450ms to 89ms (80% improvement)
- Reduced oncall incident response time by 55% by building automated debugging and self-healing systems using Prometheus metrics

Datadog | *Backend Engineer, Data Pipeline Infrastructure* 2017 - 2019

- Developed real-time log aggregation service in Python handling 2TB+ daily, reducing data processing cost per event by 64%
- Implemented automatic scaling mechanism for Kubernetes clusters, decreasing resource waste by 41% while maintaining SLA of 99.95%
- Collaborated with data science team to build anomaly detection pipeline, reducing mean-time-to-detection for system failures from 15m to 3m

PROJECTS

AsyncFlow | *Rust, Tokio, PostgreSQL, Docker* 2022 - Present

- Built distributed async task queue framework achieving 2.5M tasks/hour throughput with <100ms latency on commodity hardware
- Open-source project with 8,200+ GitHub stars, 240 contributors, used by 500+ companies processing billions of tasks monthly

VectorCache | *Go, Redis, SIMD, gRPC* 2020 - 2021

- Created in-memory vector database for ML inference applications, achieving 95th percentile latency of 2.3ms on 100M+ vectors
- Published performance benchmarks demonstrating 8.5x speedup over competitor solutions through SIMD optimization

DistributedLedger | *TypeScript, Node.js, Consensus Algorithms, Docker Compose* 2018 - 2019

- Implemented Byzantine Fault Tolerant consensus mechanism supporting 10K TPS, achieving 99.9% uptime across 50+ node test network
- Reduced validator node sync time by 73% through snapshot mechanism, enabling 15-minute node onboarding

EXTRACURRICULAR

Backend Engineering Conference Series | *Speaker & Organizing Committee Member* 2021 - Present

- Delivered 5 talks on distributed systems and microservices scaling, reaching 3,500+ engineers globally with 4.8/5.0 average rating
- Organized speaker program attracting industry leaders; conference grew from 200 to 1,200+ attendees in 2 years

Cloud Native Computing Foundation (CNCF) | *Kubernetes Working Group Contributor* 2020 - Present

- Contributed 12 merged pull requests improving performance monitoring for distributed storage, used by 400K+ deployments
- Mentored 8 open-source contributors from underrepresented communities in backend systems design

TECHNICAL SKILLS

Languages – Go, Python, TypeScript, Java, Rust
Technologies – PostgreSQL, Redis, Kafka, gRPC, GraphQL, Docker, Kubernetes, Event-driven Architecture
Tools – Prometheus, Grafana, DataDog, GitHub, Jenkins, ArgoCD, Terraform